

ODRE PQC

Installation/License Activation operating guide

For licensed customers · English · v1.2.1

Leave the app alone and connect to the server. After verifying and installing the official bundle and completing the core self-test, connect the Gateway/Core and Commercial gate to main.py. Proceed in the following order: Activate license → Start server → Verify execution status and actual business request path.

Document status: Release Candidate Customer guidance prior to final live payment, integrated bundle, obfuscation, Windows/Ubuntu clean-room verification, and artifact sealing. The final installation command and Gateway public API will be inserted after the official distribution is confirmed. This revision reflects the actual implementation of the Production CLI/Gateway provided and is not the result of connecting to a production server and performing new testing.

item	detail
official customer bundle	Commercial + Gateway + Core
Core standards	v0.2.9 (no changes)
Supported environment standards	Windows Server 2022 AMD64 / Ubuntu 24.04.4 LTS AMD64 / CPython 3.12
official website	https://pqc.odreai.com
Customer Support/Inquiries	odreai2025@gmail.com

Existing Client → Server is HTTPS/TLS, Gateway → Core is ODRE v3 protection section. It does not mean mobile-to-server E2E PQC. It requires the right distribution, support environment, permissions, and integration, and does not claim to be '100% secure' or 'unhackable'.

Quick Start after payment

First, check the Product / Plan / Units / License ID / License Key / Contract Period in the email and store the Key in a secret storage. Step 11 below is the normal customer installation sequence.

1. Verify and install the official bundle: verify the Manifest signature, then SHA-256, before installation. Check dependencies and the approved production version of the Commercial companion. (pp. 7–10)
2. Run `odre-pqc init` once, only for a new installation. Do not repeat init for updates or reinstalls that already have `config/keys/runtime/state`. (p. 10)
3. Run `odre-pqc doctor` once to diagnose the installation environment before starting the server or connecting the Gateway. (p. 11)
4. Run `odre-pqc verify` once. This is an independent Core self-test using temporary keys and a temporary DB. Do not repeat it after connecting the Gateway. (p. 11)
5. Connect Gateway/Core and the Commercial gate in `main.py`. Define the Gateway-specific identity/key/trust, protected routes, and startup/shutdown behavior. (p. 14)
6. Run `odre-pqc-commercial activate`, then authorize on the official HTTPS page with License ID, License Key, and Device Registration Code. Keep the CLI running. (p. 12)
7. Use `odre-pqc-commercial status` to confirm ACTIVE / LEASE_VALID / integrity PASS and correct Unit allocation. (p. 13)
8. Start the customer's FastAPI server and the Gateway private Core. Do not open production traffic yet. (p. 13)
9. Run `odre-pqc status` once to check authenticated runtime status of the Core that is actually running. Identify which Core was checked. (p. 13)
10. Separately verify the actual Gateway → Core → business Handler integration path. A successful Core status check does not replace this gate. (p. 15)
11. Production traffic will only be opened after all Gate PASS and operational approvals have been received.

Page order and task order are different The existing contents structure is retained. Read in this order: installation verification (pp. 10–11) → Gateway connection (p. 14) → activation (p. 12) → status checks and server startup (p. 13) → integration verification (p. 15). Stop on mismatches, unresolved requirements, or verification failures; do not bypass them.

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0. Check before starting

This is guidance for server operation and development staff of customers who have confirmed payment and received a license email. The default installation is a server-side operation and does not add PQC functionality to existing Clients.

0.1 Target and preparation

- FastAPI main.py / Customer connecting Server Gateway to an existing API.
- Windows Server 2022 AMD64 or Ubuntu 24.04.4 LTS AMD64 Manager.
- Separate installation, activation, integration, and verification times are secured. '10 minutes' is the validity time of the device registration code and does not guarantee the entire installation completion time.

0.2 Prerequisites

division	prerequisite
OS / CPU	Windows Server 2022 AMD64 or Ubuntu 24.04.4 LTS AMD64
Python	CPython 3.12 (previously audited: Windows 3.12.10 / Ubuntu 3.12.3). Final bundle verification is a separate gate.
authority	Administrator/root privileges to create installation/protection state path. The operating account has least privileges.
hour	Reliable NTP synchronization. Does not set system time back in time.
network	Official License/Activation HTTPS service outbound 443. Opening inbound port for activation is not necessary.
change safety	Secure app backup/rollback point. Do not duplicate Lease/identity/state in other installations.

Confidential information warning Do not put License Key, activation token, refresh credential, local state, or private key in chat, support ticket, source, URL, command line, shell history, or log.

1. Check payment confirmation email

After payment verification, the License ID and License Key are issued to the purchase email address. Check the Release Gate on page 27 separately for completion of final live issuance verification.

The sending address and inquiry address are different Auto Licensing Email: ODRE PQC <license@mail.odreai.com> - Outgoing only. Customer support/inquiries: odreai2025@gmail.com - Send inquiries to this address. Don't trust attachments based solely on the sending address; verify the official Manifest signature.

email entry	Customer confirmation
Product / Plan / Units	Verify that it matches the purchased product, contract plan, and required number of operational installations.
License ID / License Key	Keep both values safe. The Key is your proof of purchase and is only entered on the official HTTPS activation screen.
contract period	Check the start, end, next renewal date, and payment cycle.
Downloads / Documents	Make sure the official bundle, manifest, and signature are the same release.
Payment Reference	Masked references are for purchase confirmation purposes only. Do not use Supplier Transaction ID as a common activation input.

Which menu do you select on the homepage?

- If you received the email, prepare the installation. Complete the Core self-test and main.py connection (p. 14), then authorize under installation license activation (p. 12).
- If you did not receive the email, use 'Payment Confirmation/Send Email'. Please do not repay. Use references from your payment provider only when required by the appropriate recovery screen.

Official activation screen: <https://pqc.odreai.com/payment/register/?flow=activate>. If the issue is not resolved, cover the confidential information and contact customer support.

2. Download official product and check integrity

The Release Manifest of the official delivery channel is the standard for download file name, size, and hash. We use one customer bundle containing three components:

1. Receive the official bundle + Release Manifest + signature in the same release.
2. The manifest signature is verified using the official signature public key/trust criteria independently verified from the file being verified. Specific verification tools/commands follow final release guidance.
3. Compare the size/SHA-256 written in the manifest where the signature is valid and the actual artifact. A SHA-256 match alone does not provide issuer authentication.
4. Run the official installer only if all verification passes. Decompression, pip installation, and runtime load are not performed before verification.

Windows: Check file SHA-256

```
Get-FileHash -Algorithm SHA256 -LiteralPath "<ARTIFACT_PATH>"
```

Ubuntu: Check file SHA-256

```
sha256sum -- "<ARTIFACT_PATH>"
```

<ARTIFACT_PATH> is a placeholder, not an actual file name. Replace it with the local path to the download file specified in the manifest. The above command is for hash comparison and does not replace signature verification or installer's forced integrity verification.

historical/reference Core hash - not a bundle hash Core v0.2.9 reference SHA-256:

A8AFA10EE0AFACEF1C0FAF55FF07B96C722920E5AEE8692F6F026E804F028697

The SHA-256 of the final Commercial + Gateway + Core bundle is confirmed in the Manifest after obfuscation and sealing. Current value: Not confirmed. Do not apply this note to bundle.

3. Preparation before installation

test	PASS criteria	In case of failure
OS/CPU/Python	Exactly matches the supported combination	Installation interruption/support inquiry
system time	NTP synchronization, no significant drift	Time Cause Remedy
HTTPS egress	Official service outbound 443 available	Approval of firewall changes
Permissions/ACL	Administrator/root and restricted state paths	Permission design modifications
integrity	All hashes match the manifest signature	Discard/re-download artifact
backup	Secure application rollback point	Backup before changes

Backups may include approved copies of application code and configuration. Never copy an existing installation identity, device private key, refresh credential, or Offline Lease into a new VM/container.

4. Installation — Core + Commercial Wheel

Use the two wheels that have completed the integrity check on page 7. Just run the appropriate OS commands below. For new installations. Stop if any step fails. Gateway integration is separate; follow page 14.

4A. Windows Server 2022 AMD64 · PowerShell

```
# Run from the folder containing the two Wheel files.
# Internet connection required.
# Required Python dependencies are automatically downloaded during installation.

# 1. Check for Python 3.12
py -3.12 --version

# 2. Create a dedicated virtual environment
py -3.12 -m venv .venv

# 3. Update pip
& .\venv\Scripts\python.exe -m pip install --upgrade pip

# 4. Install Core + Commercial Wheel and required dependencies
& .\venv\Scripts\python.exe -m pip install `
    .\odre_pqc-0.2.9-py3-none-any.whl `
    .\odre_pqc_commercial-0.2.9.post3-py3-none-any.whl

# 5. Check installed dependency contracts
& .\venv\Scripts\python.exe -m pip check

# 6. Check installation version
& .\venv\Scripts\python.exe -m pip show odre-pqc
& .\venv\Scripts\python.exe -m pip show odre-pqc-commercial

# Only if this is successful will we run init in the next step.
```

4B. Ubuntu 24.04 AMD64 · Bash

```
# Run from the folder containing the two Wheel files.
# Internet connection required.
# Required Python dependencies are automatically downloaded during installation.

# 0. Install Python 3.12 virtual environment support package
sudo apt update
sudo apt install -y python3.12-venv

# 1. Check for Python 3.12
python3.12 --version

# 2. Create a dedicated virtual environment
python3.12 -m venv .venv

# 3. Update pip
.venv/bin/python -m pip install --upgrade pip

# 4. Install Core + Commercial Wheel and required dependencies
.venv/bin/python -m pip install \
    ./odre_pqc-0.2.9-py3-none-any.whl \
    ./odre_pqc_commercial-0.2.9.post3-py3-none-any.whl

# 5. Check installed dependency contracts
.venv/bin/python -m pip check

# 6. Check installation version
.venv/bin/python -m pip show odre-pqc
.venv/bin/python -m pip show odre-pqc-commercial

# Only if this is successful will we run init in the next step.
```

Updating pip is optional. pip check is a dependency check and is not a replacement for integration verification. After all installation steps are successful, proceed with `init` → `doctor` → `verify` only for new installations in the same `.venv`. Existing `config/keys/runtime/state` is preserved and `init` is not repeated when updating or reinstalling.

4. Install the official bundle

We use one official bundle that includes Commercial + Gateway + Core. Verify the manifest signature and SHA-256 before installation, and check the dependency of the approved runtime after installation. Commercial companion must match the operationally confirmed implementation/version.

Do not use arbitrary installation commands The final bundle filename, OS-specific installer/wrapper, and installation paths are defined in the official release instructions. Do not skip the installer's mandatory pre-install integrity checks or combine individual Wheels arbitrarily. See page 12 for older Commercial wrapper precautions.

4A. Windows Server 2022 AMD64

Install approved runtimes and components together with the official installer from an ACL-protected staging folder. Administrator privileges are used only for installation and initialization, and minimum privileges are granted to the operating account.

Maintain DPAPI LocalMachine protection and Administrators/SYSTEM restricted ACLs in Commercial state. Do not decrypt or copy the state file to another host.

4B. Ubuntu 24.04.4 LTS AMD64

Prepare a dedicated service account and a root-owned installation/state path. Install the Ubuntu bundle from the same release integrated into the approved CPython 3.12 runtime. The state directory is owned by root at 0700, and the encrypted state file and local state key are kept at 0600.

New installations only: initialize once

```
odre-pqc init
```

Reinitialization is not an update procedure If you have an existing config/keys/runtime/state, do not run the above command. In existing installations, updates, and reinstallations, init is not unconditionally repeated. Reruns are blocked with INITIALIZATION BLOCKED / exit code 6. To avoid this, don't delete key/state.

Installation environment diagnosis → Independent core self-test

```
odre-pqc doctor  
odre-pqc verify
```

In a normal installation, doctor and verify each run once. Next, connect main.py on page 14; do not run Core status here. Check Core status using page 13 after starting the customer server and Gateway private Core.

5. Classification of installation diagnosis, self-test, and status

Command/Gate	When do I run it?	What do I check?
Check dependency	Immediately after installing the bundle	Dependencies on approved runtimes are normal. broken by pip check requirements 0. Commercial Approved companion version matches.
odre-pqc doctor	Once before gateway connection	Installation environment including settings, key, provider, DB/permissions, etc. Diagnosis is possible even without customer main.py or the server running.
odre-pqc verify	Once, after doctor	Independent Core crypto/self-test based on temporary key/temporary DB. Not a customer FastAPI/ Gateway integration test.
Commercial activate / status	After connecting main.py, before starting the server	ACTIVE / LEASE_VALID / integrity PASS and Unit normal allocation after web approval. Not a prerequisite for core verification.
odre-pqc status	Once after server startup	Check the authenticated runtime status endpoint of the actual running core. This is not a command to read the verify result file.
Actual gateway integration verification	After starting the server and checking the runtime	Gateway → Core → Task Handler overall path and protection scope. separate final Smoke Test.

Gateway verification does not work even if you run verify again. Complete this as the independent installation self-test. Do not repeat verify after Gateway integration or Commercial activation. Verify the actual business request path on page 15.

Core status checks for evidence of running

Verifies runtime evidence such as authentication key matching, HMAC, nonce, response freshness, provider/key/security status, and worker lease. It is not a causal relationship of verify PASS → status VERIFIED/READY.

If the server does not exist or cannot authenticate, it may result in Runtime UNKNOWN / PQC PROTECTION: BLOCKED / exit code 5. Conversely, if a normal Core is running, ACTIVE / ON is possible even if verify has not been run previously, but this does not mean that the independent self-test of the normal installation flow can be omitted.

Next, connect Gateway/Core and the Commercial gate on page 14. Then proceed with activate on page 12, followed by Commercial status, server startup, and Core status on page 13.

6. Activate paid license

Run after completing Core verify and the main.py connection on page 14. The final bundle's Commercial CLI must match the approved production implementation and version.

Don't use outdated wrappers as your customer bundle standard. The old wrapper of existing C:\ODRE-PQC is an old version that asks for License ID/Key in CLI. It is not used on a final customer bundle basis. If CLI asks you to enter ID/Key, do not enter it but check that the executable file, runtime, and version match.

```
odre-pqc-commercial activate
```

Keep the install server's CLI turned on. The 8-digit registration code is valid for 10 minutes, and the 'device' is a server installation, not a mobile phone. On the official HTTPS screen, select 'Activate Installed License'.

<https://pqc.odreai.com/payment/register/?flow=activate>

License ID

00000000-0000-4000-8000-000000000000

발급 이메일에 적힌 License ID 전체를 복사하세요.

License Key

ODRE-----

표시

구매 증명인 Key는 계속 보관하세요. 이 HTTPS 승인에만 사용하며 시도 후 입력창에서 지웁니다.

장치 등록 코드

8F3K-2M9Q

1회용 코드는 10분 뒤 만료됩니다. 만료되면 명령을 다시 실행해 새 코드를 만드세요.

✓ License Key는 URL-브라우저 저장소-서버 응답-애플리케이션 로그에 남기지 않습니다. 장치 등록 코드만으로는 활성화 상태를 조회할 수 없습니다.

이 장치 승인

① License ID
Full ID of issuing email

② License Key
Purchase-proof key from the same email.

③ Device registration code
Code from the current installation server CLI

After 'Approving this device', maintain the CLI until you receive the installed private key proof
→ Unit allocation → signed offline lease.

Official public screen capture · 2026-09-08. The input is empty and the visible string is an example placeholder. This is not an actual license/code.

Key is only entered on the HTTPS screen. Approved Commercial CLI does not request or store License ID/Key. Do not put the key in the command line, environment variable, URL, history, script, ticket, or log, or export the private key outside the server.

After CLI success, check Commercial status on page 13. Do not reuse expired or failed codes; generate a new one with activate. Convert the same Trial installation to ACTIVE without deleting or reinstalling it. Reinstallation does not reset the Trial.

7. Check activation → Start server → Check runtime

1) Check commercial status

```
odre-pqc-commercial status
```

item	Check before starting server
Activation / Lease	ACTIVE / LEASE_VALID
Integrity	integrity PASS
Unit	Normal assignment to current deployment identity
Lease expiration / grace	Establish an update plan based on the actual display time. The initial open criterion is LEASE_VALID and GRACE is not used as a fallback criterion.

Do not open protected traffic without proof of activation/integrity Do not proceed unless you can prove ACTIVE / LEASE_VALID / integrity PASS or Unit allocation. It does not hide errors or remove commercial gates. There is no need to run verify again at this step.

2) Start the customer server and Gateway private core

Start the customer FastAPI server and Gateway private Core using the startup procedure defined on page 14. Keep production traffic closed. Use the approved final bundle's integration configuration for exact commands, ports, and status targets.

3) Check the actual running core status once

```
odre-pqc status
```

After clearly identifying the core to be searched, verify certified runtime evidence. If the server is absent or authentication fails, Runtime UNKNOWN / PQC PROTECTION: BLOCKED / exit code 5 may occur. This command does not display the verify result as VERIFIED/READY.

Don't use the success of other Cores instead The basic success of core status number 8000 does not mean verification of gateway private core number 18083. This port is an isolated example of the current implementation and is not a customer fixed port agreement. A private core's internal status endpoint may be disabled by default.

Do not mark the status check complete without a configured target and authentication. An approved status-check procedure for the final bundle is required. The separate Gateway private Core and actual business-path integration gate follows on page 15.

8. FastAPI / Server Gateway connection

Work location: After Core verify, before Commercial activate. The existing Android/iOS/Web Client and HTTPS API contracts will not be changed, and the existing Handler and DB will not be rewritten to suit PQC.

Existing Android / iOS / Web / other Client

↓ HTTPS / TLS

FastAPI main.py / Existing API

↓

ODRE Server Gateway / Adapter

Python PQC Client

↓ ODRE v3

ODRE PQC Core Boundary

↓

Existing Handler

Server connection item	What to confirm in official integration settings
Gateway/Core + Commercial gate	Connection of license/integrity verification of main.py integration point and protection path. Integration is not completed through install(app) or simple Core connection.
Gateway-specific identity/key / trust	Sender identity, key bound to that identity, and Core trust settings. Do not copy the secret key to the document/source/log.
protected route	Protected route list and handler bypass blocking. Does not necessarily include a public health-license control plane.
startup / shutdown	Define startup and cleanup order for the customer server and Gateway private Core, runtime-status targets, and session lifecycle. After activation, perform the actual startup on page 13.

Connect after confirming the final public API The exact import/API, configuration name, and execution command are provided as an official example of the final bundle. It is not created arbitrarily. Gateway's Python PQC Client performs ODRE v3 authentication, encryption, session, sequence, and replay verification, and the protection handler must be executed only after success.

Client → Server is HTTPS/TLS, Gateway → Core is PQC Boundary. The base product does not require a mobile SDK and does not claim mobile-to-server E2E PQC. Terminal E2E is an Optional Extension. License control traffic is separated from the business body, and customer request/response body/PQC plain text is not sent to the license service.

After configuring the connection, proceed in this order: Commercial activate (p. 12) → Commercial status, server startup, Core status (p. 13) → integration verification (p. 15).

9. Actual integrated smoke test before production

Performed after the customer FastAPI server and Gateway private core are running and the Commercial health status has been checked. Below are the passing criteria you need to check, and are not the test results that were executed and passed by creating a PDF. This is not a step to re-run verify.

Independent Gate/Test	passing criteria
Gate A: running Core status	Certified runtime evidence of the specified target core is normal. Verify Verifies execution status, not the result file.
Gate B: private Core integration	Gateway → relevant Gateway private Core → business handler actual path normal. Gate A succeeds, but does not replace the default 8000 Core state.
Existing HTTPS Client/Contract	No Client PQC header-handshake required. Maintain API request/response semantics. Actual response is normal.
Gateway-specific settings / handshake	Identity/key/trust matches, ODRE v3 handshake is normal, execution of protection handler after authentication/decryption.
session / sequence	Order normal. Explicit session re-establishment upon expiration. Deny gap/rollback.
replay / tamper / bypass	Refusal of reuse/ciphertext/header/status/body tampering. protected handler bypass 0.
License / Lease / integrity failure	NOT_ACTIVE · Invalid Lease · fail-closed on integrity failure. Protection in case of commercial gate failure route 503. Blocked outside of valid signed lease range.
Fallback / startup / shutdown	plaintext/classical fallback 0, normal start, cleanup, and re-establishment. HTTPS of the existing client is not a fallback.

Status endpoint and business integration are separate If the internal status endpoint of a gateway private core is inactive, do not substitute it for success from another core. Approved status verification procedures and actual business request path evidence for the final bundle are required, respectively. Unresolved gates are not marked as PASS.

Failure path testing can only be done in an isolated environment. replay·tamper·license-failure is performed in isolated staging/conformance. Production only confirms approved normal requests and non-destructive status. Do not bypass protection failures directly with Handler or regular communication. Open production traffic only after all Gate PASS and operational approvals have been received.

Large payload only indicates whether it is included in the final bundle and within the scope of confirmed evidence. Final product application of 18.1 MB / 32 MB byte-equivalence has not been verified in this revision. The Core single-message cap has not been removed.

10. Normal operation

Lease renewal

```
odre-pqc-commercial renew  
odre-pqc-commercial status
```

- Renewal replaces the rotating refresh credential. Reusing the previous credential may result in stale/revoked status.
- Offline Lease's `refresh\_after`, `expires\_at`, and `grace\_until` consider the status of the current installation as authority.
- The audited source uses periodic renewal and leases of up to 24 hours by default; the customer's actual signed lease values take precedence.
- A License API outage is not a payment failure. Continue only within the valid signed lease; close the protected path when its authorization can no longer be proven.
- Security patches and new releases are applied after verifying the new manifest and staging regression.

Diagnostic events cover fail-closed entry/recovery and limited installation, version, and status reasons. Do not add customer request/response bodies or PQC plaintext to logs.

11. Unit management

Under the public contract, 1 Unit means one licensed production deployment or installation. A Unit is allocated atomically to a deployment identity at activation.

situation	treatment
Multiple workers in the same installation	Since there are no public fixed formulas, check your deployment topology with support. Arbitrary sharing is prohibited due to state replication.
Separate VM/container	May count as a separate installation/deployment. Check the Unit contract before deployment.
Scale-out	Pre-check expected replica/topology and number of units
Migration	Existing installation normal deactivate → Confirm unit return → Activate new installation
Clone	Duplication of existing lease/state is prohibited. Activate with new instance identity
Unit shortage	Return existing installation or add contract unit and retry

Stale reuse of a rotating credential can reveal cloning or reuse during online renewal; do not assume offline clones are always detected immediately. Remove commercial state before imaging and activate each deployment separately.

12. Failure response

Status/Error Category	Customer Action
LICENSE_UNIT_LIMIT	Check for duplicate installation → Deactivate existing installation or expand unit
LICENSE_LEASE_EXPIRED / INVALID	Maintain blocking traffic → Check time/NTP → renew → status
STALE / revoked credential	Check state restore/cloning → stop reuse → contact support.
CLOCK_TAMPER_DETECTED	Stop time changes → restore NTP/host time → wait for support before resuming.
ARTIFACT_HASH_MISMATCH	Stop execution → Re-receive the official bundle → Manifest re-verification
INTEGRITY_CHECK_FAILED	Block protected traffic → collect doctor/verify evidence → approved reinstallation
BROKER_UNAVAILABLE	Check local license state access/authority/damage. Bypass gate prohibited
LICENSE_SERVER_UNREACHABLE	Check DNS/TLS/egress. If it is outside the effective lease range, fail-closed
UNSUPPORTED_PLATFORM	No enforcement. Go to Support Union or Contact Support

Labels may become more specific by release. Never reopen protected routes until protection evidence is restored.

13. Fail-Closed Description

Do not open a protected route unless you can prove its protection status.

- A protected request will be rejected if any of ACTIVE entitlement, integrity, or valid signed lease cannot be proven.
- Commercial gates are designed to return 503 to the protected route in case of failure.
- PQC authentication/decryption/sequence/replay verification failures must be rejected before the handler.
- Does not automatically switch to plaintext or classical-only connections after authentication failure.
- Unverified local flag, hard-coded ACTIVE, and gate removal are not used due to License service failure.

This product alone cannot establish trust in a host whose administrator/root account is fully compromised. OS hardening, account/key management, monitoring, and backup isolation remain part of the customer's operational responsibilities.

14. Backup · Transfer · Removal · Reactivation

before normal

1. Drain business traffic from existing installations and record commercial status.
2. After running `odre-pqc-commercial deactivate`, type `DEACTIVATE` exactly at the confirmation prompt.
3. Make sure to return Unit and remove local credential/private state. Do not duplicate key/state with other installations.
4. On a new server, follow the normal installation order on page 2: bundle verification/installation → new installation only init → doctor → independent verify → `main.py` Gateway/Core + Commercial gate connection.
5. After activate-web approval with new Device Code, check `ACTIVE` / `LEASE_VALID` / integrity `PASS` / Unit of Commercial status.
6. Start customer server and gateway private core → running core status → verify actual gateway integration → traffic switches after approval. After activation/connection to the gateway, verification is not performed repeatedly.

```
odre-pqc-commercial deactivate
# prompt: DEACTIVATE
```

Update/reinstall existing installation If you have an existing config/keys/runtime/state, do not reinitialize it. Rerunning init results in `INITIALIZATION BLOCKED` / exit code 6 and is not a normal update method. Follow the renewal procedure for the final, verified bundle.

If you lost your existing installation If deactivation is not possible due to abnormal loss, use our support team's Unit release/reactivation procedure. Do not bypass production by restoring Lease or private state from a backup to a new server.

15. How to request technical support

Support Channel: odreai2025@gmail.com

Information you can send	Information that should not be sent
Product/Core/Commercial Version	License Key
OS, CPU, Python version	access/refresh token, poll_token
Part of the redacted License ID and part of the Installation ID	private key, local state file
UTC time and error categories	Full URL query/header/body
Redacted result of doctor/verify/status	Customer request/response or PQC plaintext
Manifest file name and calculated artifact hash	webhook/KMS/HSM secret, dump environment variables

Before sending logs, redact email addresses, full License IDs, hostnames, internal IPs/paths, authentication headers, and token/key patterns. Attach only the minimum reproduction steps.

16. Information that must not be sent for security reasons

- License Key actual value or example resembling the actual value
- Valid values of Refresh Token, Access Token, poll_token, and Device Code
- instance/device private key, state encryption key, keystore/DPAPI export
- webhook secret, provider API secret, KMS/HSM credential
- All environment variables, settings, DB dump or internal administrator endpoint
- Actual customer identification information and application request/response body
- Internal detection thresholds and implementation details required to bypass defenses;

Do not send these values even to someone claiming to be support staff. Legitimate support never requires your raw License Key or private key.

17. Quick Installation Checklist

Each command in the normal installation flow is executed only once in that step. Do not execute the same command again just because it appears in the previous explanation. Detailed reading order follows page 2.

- [] Official bundle verification/installation: Manifest signature-SHA-256, dependency, commercial companion operation confirmed version matching
- [] Init once only for new installations (repeat for existing installations, updates, and reinstallations is prohibited)
- [] doctor 1 time PASS (before gateway connection)
- [] verify 1 time PASS (temporary key-DB-based independent Core self-test)
- [] main.py Gateway/Core + Commercial gate connection; Dedicated identity/key/trust-protection route-startup/shutdown confirmed
- [] Commercial activate + official HTTPS web authorization
- [] Commercial status: ACTIVE / LEASE_VALID / integrity PASS / Unit normal
- [] Start customer FastAPI server + Gateway private core (Production traffic is closed)
- [] Core status normal once (confirmation of actual running core and certified runtime evidence)
- [] Gateway → relevant private core → business handler actual integration verification PASS; replay/tamper/license-failure isolation verification, bypass-fallback 0
- [] Production begins after receiving all Gate PASS and operational approval.

Cases that should not be processed as completed Do not replace other Core's status success with private Gateway Core verification. If the final bundle/API/approved status confirmation procedure is unconfirmed or there is no evidence, the Production item is not checked.

18. Glossary

terminology	meaning
Server Gateway / Adapter	Server integration layer that connects existing HTTPS requests and Core Boundary. Performs ODRE v3 communication with Python PQC Client and does not require mobile SDK for basic installation.
PQC Boundary	Application Security Boundary that allows execution of business handlers through ODRE v3 authentication, encryption, session, sequence, and replay verification after the gateway.
Optional E2E Client SDK	Separate expansion for customers who require ODRE PQC end protection starting from the terminal. Not required for basic server product installation.
Payment Adapter / Billing Core	A layer that verifies and normalizes provider API, webhook, signature, and state and confirms provider-neutral Billing Event/State.
License ID / License Key	License Identifier / Secret Proof of Purchase to authorize activation.
Device Registration Code	A one-time code for the installation authorization connection displayed by the CLI. Valid for 10 minutes. Not mobile device registration.
Instance key proof	A challenge signature that proves that the installation holds the local private key.
Unit	One licensed Production distribution or installation.
Offline Lease	Installation-Signed permission belonging to the Unit and including an expiration time.
Rotating refresh credential	A secret credential that is replaced with a new value on each successful update.
Fail-Closed	A policy that closes a protection route if the protection status cannot be proven.
Core / Commercial status	Authenticated runtime evidence of the Core during execution / Activation-Permission-Lease-Integrity status. It has nothing to do with the verify result file and is verified separately.

Appendix A. Payment company independent billing contract

Payment Provider → Provider-specific Payment Adapter → ODRE normalized Billing Event / Billing State → License / Entitlement issuance → Customer activation

- Payment company information is isolated in the Payment Adapter.
- The Payment Adapter verifies provider-specific APIs, webhook signatures, and status, and normalizes them to ODRE standard Billing status.
- License/Entitlement and customer activation are not directly coupled to provider-specific transaction/subscription/customer IDs.
- Only License ID, License Key, and Device Registration Code are used for common activation after the payment completion email.
- Even if providers are added in the future, the installation, activation, and operation procedures after email will be maintained.

PayPal note — PayPal-specific payment verification and subscription data belong to the PayPal Adapter and purchase-recovery page. Customers who already received their email do not enter a PayPal subscription ID in the common activation flow.

Paddle note — Paddle transaction/subscription/customer IDs belong to the Paddle Adapter and purchase-recovery page. Customers who already received their email do not guess or enter these IDs in the common activation flow.

Appendix B. Behaviors customers should never do

- Executes when the artifact hash or manifest signature is inconsistent.
- Ignore doctor/verify failure and open protected traffic
- Change the system time to the past or bypass time checking.
- Offline Lease, commercial state, refresh credential or instance key replication
- Reuse existing lease after VM clone
- Save License Key in source code, command line, environment variable, and log.
- Run raw Wheel directly or use an unofficial mirror artifact.
- Bypass license gate or local state verification
- Connecting protected routes to classical/plaintext fallback
- ~~Forced execution on unofficial OS/runtime and marked as 'supported'~~

If any of these items occur, close traffic, revert to the last verified deployment, and investigate the cause.

Document Management and Release Gate

Document status: Release Candidate. NOT_VERIFIED means the final release proof has not been verified and does not assume PASS/FAIL. Existing audit results will not be carried over to approval for deployment of the new integrated bundle.

Gate	Status
PADDLE_LIVE	NOT_VERIFIED
LICENSE_ISSUANCE	NOT_VERIFIED
RESEND_TRANSACTIONAL_EMAIL	NOT_VERIFIED
ACTIVATION	NOT_VERIFIED
UNIT	NOT_VERIFIED
LEASE	NOT_VERIFIED
FINAL_INTEGRATED_BUNDLE	NOT_VERIFIED
FINAL_OBFUSCATION	NOT_VERIFIED
WINDOWS_CLEAN_ROOM	NOT_VERIFIED
UBUNTU_CLEAN_ROOM	NOT_VERIFIED
FINAL_ARTIFACT_SEAL	NOT_VERIFIED
PRODUCTION_RELEASE	BLOCKED
DOCUMENT_STATUS	RELEASE_CANDIDATE

Items to confirm before final distribution Finalize the bundle filename, signature trust basis, verification commands and SHA-256; OS-specific installer/wrapper and approved Commercial companion version; and the public Gateway imports/API, dedicated configuration, startup and status procedures. Separately pass the running Core status and Gateway private Core integration gates on page 15.

The command meaning and installation sequence were revised according to the Production CLI/Gateway actual implementation contract provided by the customer. While editing this PDF, the product CLI execution, server connection, and operation verification were not performed. The status value is a description of the passing criteria and is not evidence of a new PASS.

2026-09-08 Maintain a blank device registration screen capture on the public homepage. It is not an actual payment, activation, registration submission, or Live Gate verification result. The integration examples on the existing homepage are not replaced by the final Gateway API.

Product version-Core v0.2.9-Server-Gateway-Commercial source-Production will not be changed. Replacing automatic email attachments requires separate final approval. © 2026 ODRE AI · Customer Release Candidate